



UNIVERSITY OF THE PUNJAB

Roll No.

Fifth Semester 2017
Examination: B.S. 4 Years Programme

PAPER: Classical Mechanics
Course Code: PHY-301

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

Attempt all questions.

SECTION-I (Multiple Choice Questions) (10 Marks)

1. Select and tick one answer from the given multiple choice (10)
 - (i) The brachistochrone problem is the
 - (a) least area problem
 - (b) least distance problem
 - (c) least time problem
 - (d) least energy problem
 - (ii) If the Lagrangian does not involve a particular coordinate q_i such coordinate is called
 - (a) Angle coordinate.
 - (b) Fictitious coordinate
 - (c) Complete coordinate
 - (d) Cyclic coordinate.
 - (iii) The total angular momentum of a system of particles is constant if
 - (a) the total force is zero
 - (b) the total torque is zero
 - (c) the total energy is zero
 - (d) the total momentum is zero
 - (iv) Isotropy of space means that the system is invariant under
 - (a) translation in space
 - (b) rotation in space
 - (c) translation in time

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- (d) translation in space and time
- (v) Kepler's second Law of planetary motion directly follows from
- (a) Conservation of linear momentum
 - (b) Conservation of angular momentum
 - (c) Homogeneity of time
 - (d) Homogeneity of space
- (vi) Constraints that cannot be expressed as equations of coordinates and time i.e

$$f(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N, t) = 0$$

are said to be

- (a) holonomic
 - (b) Non-holonomic
 - (c) Simple
 - (d) Cruciform
- (vii) Holonomic constraints are
- (a) non-integrable
 - (b) hermitian
 - (c) noncyclic
 - (d) integrable
- (viii) The shortest distance between two points on a curved surface is
- (a) a straight line
 - (b) a tangent
 - (c) a geodesic
 - (d) a semi-circle
- (xi) The value of eccentricity ϵ for a parabola is
- (a) $\epsilon > 0$
 - (b) $\epsilon = 1$
 - (c) $\epsilon = 0$
 - (d) $0 < \epsilon < 1$
- (x) The Hamiltonian can be constructed from the Lagrangian using the formula:
- (a) $H = p_i \dot{q}_i - L$.
 - (b) $H = \dot{p}_i \dot{q}_i - L$.
 - (c) $H = \frac{\partial L}{\partial \dot{q}_i}$
 - (d) $H = \frac{1}{L}$.



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TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

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SECTION-II (20 Marks)

2. Show that the kinetic energy of a two-particle system is

$$\frac{1}{2}M \dot{R}^2 + \frac{1}{2}\mu v^2$$

where $M = m_1 + m_2$, v is the relative speed, and $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$ (reduced mass)

(5)

3. What are constraints? Distinguish between holonomic and nonholonomic constraints by giving examples.

(5)

4. Show explicitly that

$$\frac{\partial \mathbf{x}}{\partial q_i} = \frac{\partial \dot{\mathbf{x}}}{\partial \dot{q}_i}$$

where $\mathbf{x} = \mathbf{x}(q_1, q_2, \dots, q_n)$. (5)

5. Show that when a particle describes a circle under a central attraction directed to a point in the circumference, the law of force is the inverse fifth power of the distance. (5)

SECTION-III (30 Marks)

6. (a) Show that the equation of a curve for which surface area is minimum is a catenary

$$x = a \cosh \frac{y-b}{a}$$

where a and b are constants.

- (b) Use Hamilton's principle of least action to derive Hamilton's equations of motion

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

(5+5)

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7. (a) Use variational method to derive Euler-Lagrange equation of motion for a given functional

$$I = \int_{x_1}^{x_2} f[y(x), y'(x), x] dx.$$

- (b) Show that the transformation

$$\begin{aligned} P &= q \cot p, \\ Q &= \log \left(\frac{\sin p}{q} \right), \end{aligned}$$

is canonical (5+5)

8. (a) Show that, if a transformation from (q, p) to (Q, P) be canonical then the bilinear form

$$\sum_i (\delta p_i dq_i - \delta q_i dp_i),$$

is invariant under the canonical transformation.

- (b) A particle of mass m glides without friction on a cycloid, given by

$$\begin{aligned} x &= a(\theta - \sin \theta) \\ y &= a(1 + \cos \theta), \end{aligned} \quad 0 \leq \theta \leq 2\pi$$

Find the Lagrange's equation of motion. (5+5)





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Roll No.

Fifth Semester 2017
Examination: B.S. 4 Years Programme

PAPER: Mathematical Methods of Physics-I
Course Code: PHY-302

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

Instructions. Attempt all questions

Section-I

Fill in the blank or answer true/false.

1. If f is a scalar function with continuous second partial derivatives, then $\nabla \times (\nabla f) = \text{curl}(\text{grad } f) =$
2. If $\mathbf{F}$ is a vector field having continuous second partial derivatives, then $\nabla \cdot (\nabla \times \mathbf{F}) = \text{div}(\text{curl } \mathbf{F}) =$
3. Suppose that C is a piecewise-smooth simple closed curve bounding a simply connected region R . If $P, Q, \frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$ are continuous on R , then $\oint_C P dx + Q dy =$
4. $g^{ij} g_{jk} = \delta_k^j$ (True/False)
5. $\nabla \phi(r) = \frac{d\phi}{dr} \hat{r}$ (True/False)
6. $e^i = g^{ij} e_j$ and $e_i = g_{ij} e^j$ (True/False)
7. The function $f(z) = e^{\frac{1}{z-1}}$ has an essential singularity at $z = 1$ (True/False)
8. If $z = 3 + 4i$, then $\text{Re}\left(\frac{z}{\bar{z}}\right) = \text{Re}(z/\bar{z}) =$ and $\text{Im}\left(\frac{z}{\bar{z}}\right) = \text{Im}(z/\bar{z}) =$
9. If $f(z) = e^z$ is periodic with period
10. The only possible singularities of a rational function are poles (True/False)





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Fifth Semester 2017
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Roll No.

PAPER: Mathematical Methods of Physics-I
Course Code: PHY-302

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.
SUBJECTIVE TYPE

Section-II (Short Questions)

Marks=20

1. If $\mathbf{B} = \nabla \times \mathbf{A}$, show that

$$\oint \mathbf{B} \cdot d\boldsymbol{\sigma} = 0,$$

for any closed surface S .

2. Expand

$$f(z) = \frac{1}{z(1-z)^2}$$

in a Laurent series valid for $0 < |z| < 1$.

3. Determine whether $z = 0$ is an isolated or non-isolated singularity of $f(z) = \tan(1/z)$.
4. A vector $\mathbf{A}$ is decomposed into a radial vector $\mathbf{A}_r$ and a tangential vector $\mathbf{A}_t$. If $\hat{\mathbf{r}}$ is a unit vector in the radial direction, show that

$$\begin{aligned} \mathbf{A}_r &= \hat{\mathbf{r}}(\mathbf{A} \cdot \hat{\mathbf{r}}), \\ \mathbf{A}_t &= -\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{A}). \end{aligned}$$

5. Evaluate $\oint_C \frac{z+1}{z^4+4z^3} dz$, where C is the circle $|z| = 1$.

Section-III

Marks=30

1. Show that

$$\int_0^\pi \frac{d\theta}{(a + \cos \theta)^2} = \frac{a\pi}{(\sqrt{a^2 - 1})^3}, \quad (a > 1).$$

2. Use Cauchy's residue theorem to evaluate

$$\oint_{|z|=\frac{1}{2}} \frac{\cot \pi z}{z^2} dz.$$

3. Suppose $A = [a_{ij}]$, $B = [b^{ij}]$ and that $B = A^{-1}$. By considering the determinant $a = |A|$, show that

$$\frac{\partial a}{\partial u^k} = a b^{ji} \frac{\partial a_{ij}}{\partial u^k}.$$

4. We may define Christoffel symbols of the first kind by

$$\Gamma_{ijk} = g_{il} \Gamma_{jk}^l.$$

Show that these are given by

$$\Gamma_{kij} = \frac{1}{2} \left\{ \frac{\partial g_{ik}}{\partial u^j} + \frac{\partial g_{jk}}{\partial u^i} - \frac{\partial g_{ij}}{\partial u^k} \right\}.$$

5. Prove the expression $\nabla \cdot \mathbf{a} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 a_1) + \frac{\partial}{\partial u_2} (h_1 h_3 a_2) + \frac{\partial}{\partial u_3} (h_1 h_2 a_3) \right]$ in orthogonal curvilinear coordinates.



UNIVERSITY OF THE PUNJAB

Roll No.

Fifth Semester 2017
Examination: B.S. 4 Years Programme

PAPER: Solid State Physics-1
Course Code: PHY-303

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE

Q.1 Encircle the correct answer (from the given multiple choices) in each part. (1×10 = 10)

- A) How many total number of lattice types are there in three dimensions?
i) 5 ii) 8 iii) 14 iv) 16
- B) Which of the following combination of structures has similar packing fraction?
i) FCC and SC ii) BCC and SC iii) BCC and HCP iv) HCP and FCC
- C) A primitive cell has lattice points per unit cell
i) One ii) Two iii) Three iv) Four
- D) Reciprocal of FCC lattice is
i) FCC lattice ii) BCC lattice iii) SC lattice iv) HCP lattice v) None of these
- E) Optical phonon branch appears in
i) monoatomic lattice ii) diatomic lattice iii) triatomic lattice iv) None of these
- F) According to Dulong-Petit law, the phonon heat capacity C_v
i) strongly depends on temperature ii) independent of temperature iii) remains constant at all temperatures iv) ii) and iii) v) none of these
- G) Van der Waals interactions in inert gas crystals are always
i) repulsive ii) attractive iii) neither attractive nor repulsive iv) zero v) none of these
- H) At high temperatures, phonon heat capacity, C_v (according of Debye model) varies as:
i) T^3 ii) $T^{3/2}$ iii) T^2 iv) T v) None of these
- I) In monatomic lattice, the frequency of the wave at long wavelengths varies with k as:
i) k ii) k^2 iii) k^3 iv) independent of wave-vector k
- J) Bulk modulus (B) and compressibility (K) are related to each other as:
i) B has direct relation to K ii) B has inverse relation to K iii) B is not related to K iv) None of these

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UNIVERSITY OF THE PUNJAB

Fifth Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Solid State Physics-1
Course Code: PHY-303

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Note: Attempt all questions. Write to the point answer of theoretical part of each question.

Q.2 Give to the point answer / short description of each question. (4 × 5 = 20)

- Draw (211), (200), (101) and (111) crystallographic planes in cubic unit cell.
- Describe briefly what is meant by anharmonic crystal interactions.
- Write down the steps to find Miller indices of a plane whose intercepts are 2, 1, 2.
- How an inert gas crystal holds together? Discuss the origin of Van der Waals interactions in crystals?
- For hexagonal close-packed structure (HCP), show that the $c/a = 1.633a$.

Q.3

- Derive the dispersion relation for a diatomic linear chain of atoms with masses M_1 and M_2 ($M_1 > M_2$) by taking into account nearest neighbor interactions. (7)
- Give physical significance of the appearance of optical and acoustical branches in the dispersion relation. Why frequency gap between two modes occurs in case of diatomic lattice? (3)

Q.4

- Derive the diffraction condition in reciprocal lattice. (7)
- What is Madelung constant? Find its value for an infinite one dimensional array of alternative ions. (3)

Q.5

- What are the discrepancies in the classical theory of lattice heat capacity of solids? Derive an expression for lattice heat capacity of solids on the basis of Einstein model. (3+4)
- Discuss how Einstein model fits well with the experimental observations for low and high temperature limits. (3)

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Roll No.

Fifth Semester 2017
Examination: B.S. 4 Years Programme

PAPER: Electronic Devices and Circuits
Course Code: PHY-304

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

Q.1 Mark the Correct one (OBJECTIVE)

10

1. The efficiency of transformer coupled Class –A power Amplifier is
a) 25% b) 50% c) 79 % d) 100%
2. Pie (π) Filter circuit consists of _____ and two Capacitors.
a) Resistor b) Inductor c) Battery d) transistor
3. In the forward blocking region the SCR is:
a) Reverse biased b) in the off state c) in the on state d) at the point of break down
4. Which of the following is not a characteristic of the UJT?
a) intrinsic standoff ratio b) negative resistance
c) peak point voltage d) bilateral conduction
5. Flip- Flop is a circuit which can store:
a) two bit memory b) one bit memory c) both a, and b
d) non of them
6. UJT transistor use in:
a) Oscillator b) Phase control c) Amplifier d) both a and b
7. A diode that has a negative resistance characteristic is the:
a) Schottky diode b) tunnel diode c) laser diode d) Diac
8. An LED
a) emits light when reverse biased b) senses light when reverse biased
c) emits light when forward biased d) produces light with longer wavelength
9. The FET is:
a) a unipolar device b) voltage controlled device
c) current controlled device d) both a & c
10. The Miller input capacitance of an amplifier is dependent in part on
a) the input coupling capacitor b) the voltage gain
c) the bypass capacitor d) non of these



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Course Code: PHY-304

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Attempt all Questions

Q.2 Short Answer

5* 4= 20

1. What is a π Filter? Why Bridge rectifier is preferable to a two diodes full-wave rectifier?
2. What is the meaning of feedback and what is the difference of positive and negative feedback?
3. Discuss basic idea of Light Emitting Diode & LCD.
4. What is the Low frequency response in BJT?
5. What is a Miller effect?

Q.3

a) Draw the circuit of a Bi-stable Multivibrator. Explain its operation, where it use, what type of output it produce. (7)

b) Write down the Characteristics of OP Amp. (3)

Q.4

Explain and analyze of Push pull class AB amplifier? Calculate the maximum output power and its efficiency. (10)

Q.5

a) What type of feedback does a Oscillator circuit require? And what are the condition for oscillator circuits. (3)

b) Explain the function of Phase shift Oscillator. Design it and calculate its frequency. (7)

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Roll No.

Fifth Semester 2017

Examination: B.S. 4 Years Programme

PAPER: Quantum Mechanics-I

TIME ALLOWED: 30 mins.

Course Code: PHY-305

MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

SECTION – I (OBJECTIVE TYPE)

Q1: Choose (encircle) the best possible answer from the given. (1x10=10)

1. Which of the following is a non-linear operator?
 - a) Differential Operator
 - b) Square Operator
 - c) Momentum Operator
 - d) Position Operator
2. If $[\hat{A}, \hat{B}] = 0$, then both operators can be determined:
 - a) Simultaneously
 - b) Precisely
 - c) Both a & b
 - d) Impossible to determine
3. $[\hat{L}_y, \hat{p}_y] =$
 - a) $\hat{L}_y$
 - b) $\hat{L}_x$
 - c) $\hbar \hat{L}_z$
 - d) Zero
4. The Eigen values of a Hermitian operator are always
 - a) Non-imaginary
 - b) Positive numbers
 - c) Real
 - d) All above true

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Fifth Semester 2017

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Quantum Mechanics-I
Course Code: PHY-305

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q.2 Give short answers to the following questions. (4x5=20)

- Define Hermitian operator and prove that $\hat{x}$ is hermitian.
- Show that the Eigen value spectrum of $\hat{L}_x$ is $m\hbar$. Where m has integer values.
- Describe the terms briefly, complete set of linear independent objects, degenerate Eigen values.
- Briefly explain the physical interpretation of ladder operator in Quantum mechanics.
- What is a zero point energy? If a classical oscillator has energy $\frac{1}{2}\hbar\omega$, what is its amplitude?

Q.3(a) State and prove uncertainty principle in Quantum mechanics.

(b) Show that $(\Delta A)^2$ is always real number. (6+4)

Q.4 (a) Write down the expressions for the angular momentum operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ in Cartesian coordinates and convert them into spherical polar coordinates.

(b) Also calculate eigen value spectrum of $\hat{L}_z$. (6+4)

Q.5 (a) State and prove Ehrenfest's Theorem. (2+5,3)

(b) Write down three canonical conjugate observable.

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